

Career Opportunities

اس باب سے جڑے پیشے ہیں: مشین لرننگ انجینئر، سائبر سیکیورٹی اینالسٹ، اے آئی محقق، ڈیٹا پرائیویسی آفیسر اور اخلاقی ہیکر۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 17 of 17

Career Opportunities

LEARN IT

This unit covers neural networks, deep learning, secure collaboration, data protection and digital equity. Mastery of these ideas opens high-demand careers in **Artificial Intelligence** and **Cybersecurity** — including in Pakistan's banks, software houses, universities and public projects linked to the National AI Policy 2025.

Career	What the person does
Machine Learning Engineer	Builds and trains machine-learning and neural-network models used in face recognition, NLP and recommendation systems.
Cybersecurity Analyst	Monitors systems for threats such as malware and phishing, and applies mitigation (firewalls, 2FA) to protect organisational data.
AI Researcher	Studies and develops new techniques in artificial intelligence and deep learning, often in university labs or high-tech R&D centres.
Data Privacy Officer	Ensures the organisation's data handling complies with legal privacy principles, protecting sensitive user information and secure collaboration.
Ethical Hacker	Legally tests systems and networks for vulnerabilities (penetration testing) before malicious attackers can exploit them via DoS or malware.

WORKED EXAMPLE

Worked example: match the job.

1. "Train a model that tags Urdu news headlines." → **Machine Learning Engineer**.
2. "Watch the bank's logs for phishing domains and turn on 2FA." → **Cybersecurity Analyst**.
3. "With written permission, try to break the college portal before criminals do." → **Ethical Hacker**.

REMEMBER IT

AI careers lean on Python, maths and ethics. Cyber careers lean on threats, 2FA, firewalls and careful testing. Both need clear communication.

COMMON MISTAKE

Mistake: Confusing an ethical hacker with a criminal hacker.

Fix: An ethical hacker has **permission** and reports weaknesses so they can be fixed. Attacking without permission is a crime.

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CHECK YOURSELF

1. Who legally tests a network for weaknesses before attackers do?
2. Which career focuses on compliance and protecting sensitive user information?

Answers: 1. Ethical hacker. 2. Data Privacy Officer.

Next lecture: Chapter review (in the chapter lecture PDF)